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ORIGINAL RESEARCH ARTICLE

"Can We Trust Them?" An Expert Evaluation of Large Language Models to Provide Sleep and Jet Lag Recommendations for Athletes

Jacopo Vitale[1,2] · Alan McCall[3,4,5] · Andrea Cina[1,6] · Athlete Travel, Sleep Interest Group (ATSIG) · Dina C. Janse van Rensburg[7,8] · Shona Halson[9,10]

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Abstract

Background With the increasing use of artificial intelligence in healthcare and sports science, large language models (LLMs) are being explored as tools for delivering personalized, evidence-based guidance to athletes.

Objective This study evaluated the capabilities of LLMs (ChatGPT-3.5, ChatGPT-4, and Google Bard) to deliver evidence-based advice on sleep and jet lag for athletes.

Methods Conducted in two phases between January and June 2024, the study first identified ten frequently asked questions on these topics with input from experts and LLMs. In the second phase, 20 experts (mean age 43.9 ± 9.0 years; ten females, ten males) assessed LLM responses using Google Forms surveys administered at two intervals (T1 and T2). Inter-rater reliability was evaluated using Fleiss' Kappa, and intra-rater agreement using the Jaccard Similarity Index (JSI), and content validity through the content validity ratio (CVR). Differences among LLMs were analyzed using Friedman and Chi-square tests.

Results Experts' response rates were high (100% at T1 and 95% at T2). Inter-rater reliability was minimal (Fleiss' Kappa: 0.21–0.39), while intra-rater agreement was high, with 53% of experts achieving a JSI ≥ 0.75. ChatGPT-4 had the highest CVR for sleep (0.67) and was the only model with a valid CVR for jet lag (0.68). Google Bard showed the lowest CVR for jet lag (0%), with significant differences compared to ChatGPT-3.5 ($p = 0.0073$) and ChatGPT-4 ($p < 0.0001$). Reasons for inappropriate responses varied significantly for jet lag ($p < 0.0001$), with Google Bard criticized for insufficient information and frequent errors. ChatGPT-4 outperformed other models overall.

Conclusions This study highlights the potential of LLMs, particularly ChatGPT-4, to provide evidence-based advice on sleep but underscores the need for improved accuracy and validation for jet lag recommendations.

Key Points

- ChatGPT-4 consistently delivered more accurate and professional answers, particularly in addressing sleep-related questions, outperforming both ChatGPT-3.5 and Google Bard.
- While large language models (LLMs) performed well on sleep-related queries, they struggled with more complex topics like jet lag, highlighting the need for further development and refinement.
- LLMs can enhance information accessibility and decision-making in sports science but must be used as part of an expert-driven process to ensure accuracy and consistency.

Dina C. Janse van Rensburg and Shona Halson contributed equally to this work and share senior authorship.

The members of "Athlete Travel and Sleep Interest Group (ATSIG)" are listed in Acknowledgements section.

Extended author information available on the last page of the article

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1 Introduction

The advent of natural language processing (NLP), a pivotal branch of artificial intelligence (AI), has marked a transformative shift in how we interact with technology and information. NLP focuses on the understanding and generation of human language, enabling AI systems to interpret, process, and produce text in a way that mimics human communication [1]. One of the most notable developments in this field is ChatGPT (Chat Generative Pre-trained Transformer), which OpenAI introduced in November 2022. Unlike traditional AI systems, ChatGPT is not only designed to provide accurate predictions and align with human expectations in its responses but also to facilitate meaningful interactions, adapt to diverse contexts, and assist in complex problem-solving with a conversational approach. This innovation has been revolutionary, particularly with ChatGPT's ability to interact conversationally and handle complex queries with an impression of understanding and coherence [2]. ChatGPT has quickly garnered significant attention, accumulating over one million users within days of its release [3]. Since then, chatbots have been rapidly integrated into both public and academic spheres [2].

The integration of ChatGPT into humans' daily lives is not without concerns. Issues such as ethics, transparency, potential biases, and the risk of generating inaccurate or misleading information, commonly called "hallucination", remain of concern [4]. Studies have highlighted discrepancies in the accuracy and reliability of responses provided by ChatGPT, particularly in specialized fields such as orthopaedic and musculoskeletal sciences [5–7], clinical radiology [8], fertility [9], and cancer [10]. These findings call for a cautious and critical approach when using ChatGPT and other chatbots for clinical and sports-related advice.

ChatGPT version 4.0, released by OpenAI on 14 March 2023, introduced further improvements, including a better grasp of nuanced prompts and more context-aware responses. This version also introduced a paid subscription model to access its advanced features [11]. Previous studies have indicated that the paid ChatGPT-4 performs significantly better than ChatGPT-3.5 in providing accurate information across various fields, such as hyperlipidemia [12], breast cancer [13] and myopia care [14]. However, there is still a lack of comparative studies on the free and paid versions of ChatGPT and other chatbots (e.g., Google Bard or Claude) in the context of sports science information.

Athletes, coaches and sport science and sports medicine practitioners often seek advice on optimising performance, recovery, and managing issues related to their daily practice. Practitioners report using various sources of evidence to guide their decisions and practices with athletes [15, 16]. For instance, the most frequently mentioned sources of evidence for mental fatigue and recovery included colleagues, peer-reviewed research, podcasts, websites and blogs, as well as attendance at conferences and seminars [16]. Similarly, peer-reviewed research was highly ranked by US sport practitioners as a source of information for planning their training and recovery programs, followed by conferences and seminars [15]. In the field of sports science, the potential for AI tools to provide evidence-based information seems to be particularly intriguing and promising. Given the complexity of topics like sleep and jet lag and their impact on athletic performance, the ability of large language models (LLMs) to deliver appropriate, relevant and practical insights could represent a significant advancement. While both reduced sleep and jet lag have been shown to negatively affect performance [17], the strength and clarity of evidence vary depending on the specific aspect under investigation. Sleep restriction and poor sleep quality have been associated with impaired reaction time, reduced sport-specific performance, and diminished cognitive performance in athletes [18]. Similarly, jet lag—resulting from rapid travel across multiple time zones—can disrupt circadian rhythms, alter hormonal profiles, and impair both physical and mental performance for several days post-travel [17]. However, due to the multifactorial nature of these conditions and individual variability among athletes, not all effects are consistently observed across studies. As AI systems are increasingly called upon to support sports science, evaluating their effectiveness in providing valid information and recommendations becomes essential.

To our knowledge, this is the first study to assess how well several LLMs perform in delivering evidence-based information and recommendations on sleep and jet lag for athletes. Coaches and sports science staff are often responsible for helping athletes implement effective strategies in these areas, making access to accurate, evidence-based guidance especially important. The types of recommendations typically sought include sleep quantity and quality guidelines, identification of sleep disruptors, use of wearables, pre-sleep routines, napping strategies, caffeine limits, as well as evidence-based approaches for managing jet lag—such as circadian timing, travel direction, resynchronization timelines, melatonin use, and the role of chronotype. By analysing LLM responses to common questions in these areas, we aimed to determine the extent to which these AI tools can assist coaches, sports science and sports medicine practitioners in managing challenges associated with sleep and jet lag. Specifically, this study aimed to evaluate: (1) the content validity of ChatGPT-3.5, ChatGPT-4 and Google Bard, in providing appropriate information on sleep and jet jag for

athletes and coaches; (2) the differences in performance and characteristics between the three LLMs; and (3) inter-rater reliability and intra-rater agreement in a group of 20 experts evaluating the same content generated by the three LLMs on these topics. We hypothesised that we would observe: (1) good content validity of the LLMs for both the sleep and jet lag topics; (2) better performances of ChatGPT-4 compared to ChatGPT-3.5 and Google Bard; and (3) weak inter-rater reliability but high intra-rater agreement.

2 Methods

2.1 Study Design

This study was conducted in two phases, between January and June 2024. The first phase involved the following preliminary methodological steps: (1) identifying ten relevant frequently asked questions (FAQs) on the topics of sleep and jet lag in athletes; (2) recruiting a multidisciplinary group of 20 experts; and (3) conducting a pilot test with English-speaking individuals to assess the understandability, formatting, readability, and clarity of the online survey used by the experts to evaluate the LLMs. The second phase involved the experts' assessment of the LLMs. This evaluation process was conducted through online Google Forms surveys and was carried out twice, with an 8-week interval, to assess inter-rater and intra-rater reliability. As officially specified by a letter from the local Ethical Committee of the Principal Investigator (dated 15 December 2023), ethics approval was not required for the present study because the research project does not fall within the scope of the Human Research Act [19]. To ensure high-quality reporting standards, the reporting guidelines for the early-stage clinical evaluation of decision support systems driven by AI (DECIDE-AI) were followed [20].

2.2 Identification of Frequently Asked Questions (FAQs)

The typical FAQs posed by athletes, coaches and sports medical staff during consultations, presentations or meetings, on the general topics of "sleep in athletes" and "jet lag in athletes", were first individually collected and then shared among the leading investigators of this study (JV, SH and DVR). Additionally, ChatGPT-4 was prompted with the following question: "Please create a list of the ten most frequently asked questions by coaches and medical staff on the topic of sleep in athletes". The same prompt was used for jet lag. These questions reflected ChatGPT-4's training data and not actual queries entered by coaches or staff. Finally, the ten most relevant FAQs on these topics were reviewed and identified by the same three lead investigators (Table 1).

Table 1 The ten frequently asked questions (FAQs) on sleep and jet lag, based on expert experience and ChatGPT-4 outputs

Sleep	
Q1	How much sleep should an athlete obtain per night?
Q2	Which are the key variables negatively influencing sleep in athletes?
Q3	What are the potential consequences of inadequate sleep on athletic performance?
Q4	Which is the best method to assess sleep in athletes?
Q5	Should wearable devices be used in athletes to assess their sleep?
Q6	Does sleep play a role in injury risk in athletes?
Q7	What are effective pre-sleep routines or rituals that athletes can adopt to improve sleep onset and overall sleep quality?
Q8	Is daytime napping beneficial for athletes?
Q9	What impact does late-night training or competition have on sleep?
Q10	What is the daily limit of caffeine for athletes?
Jet lag	
Q1	What are the differences between travel fatigue and jet lag?
Q2	What is the best method to assess jet lag in athletes?
Q3	What is the best method to assess travel fatigue in athletes?
Q4	Why is travel direction important for travelling athletes?
Q5	How long does it take to resynchronise an athlete's body clock after long-haul travel?
Q6	What strategies can athletes use to minimize the effects of jet lag before, during, and after travel?
Q7	Does jet lag have a significant negative impact on athletes' sleep following long-distance travel?
Q8	Does jet lag have a significant negative impact on athletes' performance following long-distance travel?
Q9	What role do sleep aids or melatonin supplementation play in managing jet lag?
Q10	Does the athletes' chronotype have a role in resynchronising after long-haul travels?

2.3 The Group of Experts

The lead investigators identified and recruited a multidisciplinary group of 20 experts. The expert panel, defined as the Athlete Travel and Sleep Interest Group (ATSIG), comprised scientists, researchers, clinicians, professors and scientific consultants with diverse backgrounds (e.g., sleep science, sports science, sports nutrition, sports medicine and sports psychology). Experts were selected by the three lead investigators based on their scientific track record (e.g., peer-reviewed publications relevant to sleep and jet lag in athletes), clinical experience working with elite athletes in these areas, or both. Once agreement was reached on the final panel, each expert was contacted individually via email. They received a document outlining the study protocol, including the study aims, procedures, their expected involvement, and required tasks. All experts voluntarily agreed to participate, and no incentives were provided to any of the investigators for their participation. The ATSIG was sex-balanced (ten females and ten males), mainly composed of researchers/scientists (55%, 11/20), with a mean age of 43.9 ± 9.0 years with a mean of 13.0 ± 4.5 years of experience in the research fields of sleep and jet lag. Experts affiliated with institutions in multiple countries ensured a range of international perspectives (Table 2).

Table 2 Characteristics of the Athlete Travel and Sleep Interest Group (ATSIG)

Variable	Descriptive statistics	Value
Female sex	(*n*, %)	*n* = 10, 50.0%
Age (years)	Mean ± SD	43.9 ± 9.0
	Range (min–max)	28.0–61.0
	95% CI (lower–upper)	39.8–48.0
Years of experience (years)	Mean ± SD	13.0 ± 4.5
	Range (min–max)	6.0–20.0
	95% CI (lower–upper)	10.9–15.1
Main profession(s) (*n*, %)[a]	Researcher/scientist	11, 55%
	Professor (academia)	4, 20%
	Scientific consultant	5, 25%
	Medical doctor/physician	4, 20%
Nationality (*n*, %)	Italy	1, 5%
	France	1, 5%
	Germany	1, 5%
	England	1, 5%
	Ireland	2, 10%
	Poland	1, 5%
	Spain	1, 5%
	Japan	1, 5%
	Canda	1, 5%
	South Africa	2, 10%
	Australia	6, 30%
	USA	2, 10%

Values are presented as means ± standard deviations, range (minimum-maximum), 95% confidence interval, and as percentages (%)

[a]More than one option was possible

2.4 Pilot Testing

Before the ATSIG members assessed the LLMs, a pilot test was conducted to evaluate the online survey's understandability, formatting, structure, readability and clarity. Eight English-speaking researchers from biomedical fields of research (excluding sports science) completed the survey on sleep, providing verbal feedback to the lead investigator (JV). These researchers were initially contacted by e-mail and then recruited by the lead investigator at his own institution using convenience sampling based on professional networks and availability. After two rounds of revisions, primarily focusing on the wording of the questions and the structure of the Likert scale (described below) for evaluating the appropriateness of the LLMs' responses, the three lead investigators approved the final version of the online survey.

2.5 Large Language Models (LLMs)

The LLMs utilised for the study were ChatGPT-3.5, ChatGPT-4, and Google Bard. ChatGPT-4 require a subscription fee, whereas Google Bard and ChatGPT-3.5 are available for free. The subscription to GPT-4.0 was purchased by the study's lead investigator specifically for the purposes of this study, using institutional funds. The ten FAQs were submitted to these publicly accessible LLMs through their online portals on the same day, for both surveys (25 January 2024). The following text was included for each FAQ: "Act as an expert scientist on this topic which is up to date with the latest scientific research and has years of experience counseling athletes and coaches with empathy and clarity. Provide comprehensive and easily understandable answers and limit your answer to 150 words". After generating the answers, an investigator (JV) copied and pasted them in a Word file and the online Google Forms survey.

2.6 Online Surveys

The members of the ATSIG received the answer sets on sleep and jet lag provided by the three LLMs and rated them using the free online application Google Forms. The online surveys were divided into six sections: (1) Instructions, (2) General information and anonymization, (3) Answer set #1 (first LLM), (4) Answer set #2 (second LLM), (5) Answer set #3 (third LLM), (6) Final evaluation. For both surveys, the order of presentation of the LLMs to the experts was randomized, and raters were blinded to LLMs. This evaluation process was carried out twice: at baseline (T1) and 8 weeks later (T2). Further, to avoid overburdening the experts, the two surveys at T1 and T2 were administered 2 weeks apart

from each other (T1: sleep on 26 February 2024 and jet lag on 12 March 2024; T2: sleep on 29 April 2024 and jet lag on 13 May 2024).

2.7 Rating System

Each response by the LLMs was subjected to an evaluation by the ATSIG experts answering the following question: "*To what extent do you agree that the information provided in the text is appropriate?"* The answer options were presented on a symmetrical and balanced horizontal 6-point Likert scale with extremes ranging from "Strongly disagree" (1) to "Strongly agree" (6) (see Online Supplemental Material (OSM) Resource 1).

In case of partial or total disagreement with the answers, the ATSIG members were asked to identify the reason from one of these categories: (1) "Off topic, the answer is not pertinent to the question", (2) "Clear mistakes in the answer", (3) "Too much information, not all necessary", (4)"Too few information, not enough for an exhaustive answer", (5) "Language issues, not suitable for athletes or coaches", and (6) "Other reasons". If "Other reasons" was selected, the experts could briefly describe their rationale in an open-text comment field, although this was not mandatory.

At the end of each set of answers, the experts were asked to evaluate, using the same Likert scale, to what extent they agreed that the information provided by each LLM was understandable, clear, professional and of appropriate length. In the last section of the survey, the ATSIG members were presented with four additional questions. These questions aimed to determine their preference for the best LLM and to gather their general perspective and expectations on using AI tools in sport and sport science. The inquiries and questions included at the end of each set of answers and at the end of the survey are reported as OSM Resource 2.

2.8 Statistical Analysis

Ordinal and non-normally distributed data are reported as median and interquartile range (IQR), normally distributed continuous data are reported as mean ± standard deviation (SD), and categorical variables are reported as percentages (%).

2.8.1 Reliability and Agreement

For the inter-rater reliability, the Fleiss' kappa was calculated six times (3 LLMs × 2 surveys) and interpreted as none (0–0.20), minimal (0.21–0.39), weak (0.40–0.59), moderate (0.60–0.79), strong (0.80–0.90), and almost perfect (> 0.91) [21, 22]. For the intra-rater agreement, the Jaccard Similarity Index (JSI) was calculated 120 times (20 raters × 3 LLMs × 2 surveys) and interpreted as low (0–0.25), moderate (0.26–0.50), high (0.51–0.75), and very high (0.76–1.0) [23].

2.8.2 Content Validity

LLMs' content validity is assessed using the content validity ratio (CVR). The CVR is used to quantify the degree of agreement among experts on the appropriateness of the answers by the LLMs and it is calculated as:

$$\mathrm{CVR} = \frac{\left(n\mathrm{e} - \left(\frac{N}{2}\right)\right)}{\left(\frac{N}{2}\right)}$$

where n_e is the number of experts indicating an item "appropriate" and N is the total number of experts. Therefore, considering the sample size of experts ($n = 20$) and the recognized Lawshe's CVR critical values [24], a CVR minimum value of 0.50 is expected for each answer to be considered valid (corresponding to a 75% agreement). Specifically, CVR was calculated by converting the 6-point Likert scale into a binary decision as follows: scores from 1 to 3 were classified as "non-appropriate" and scores from 4 to 6 as "appropriate" (CVR1). To be more conservative, a second CVR was also calculated including only scores of 5 and 6 as "appropriate" (CVR2). The Content Validity Index (CVI) was calculated as the average of the CVRs for the ten answers. In addition, to test the differences in CVRs among LLMs and between surveys, a two-way analysis of variance (ANOVA) with Bonferroni's multiple comparisons test was applied, as the data were normally distributed, confirmed by the Kolmogorov–Smirnov test. CVR1 was used for the analysis, whereas CVR2 was used for descriptive purposes.

2.8.3 Differences Among LLMs and Between Surveys

The normality of the distribution of raters' scores was separately checked using the Shapiro–Wilk test for each question and each LLM for both surveys. Given the non-normal distribution of data and to be more conservative, the non-parametric Friedman test followed by Dunn's multiple comparisons was applied to assess the differences in raters' scores among ChatGPT-3.5, Google Bard and ChatGPT-4. The same statistical approach was adopted to evaluate the differences in answers towards understandability, clarity, professionality and length among the LLMs. Chi-square tests (χ^2) were applied to examine differences in the reasons for inappropriate responses among ChatGPT-3.5, Google Bard and ChatGPT-4 for the sleep and jet lag surveys separately. The differences in the experts' expectations and attitudes toward LLMs between the sleep and the jet lag surveys were assessed with the non-parametric Wilcoxon test.

Effect sizes (ES) for significant pairwise comparisons were calculated using Cohen's d ($(M_2 - M_1)/SD_{pooled}$) for parametric comparisons and the r value ($r = Z/\sqrt{N}$) for non-parametric comparisons. ES were considered to be trivial (< 0.20), small (0.21–0.60), moderate (0.61–1.20), large (1.21–2.00) or very large (> 2.00) [25]. Statistical analyses were performed using GraphPad Prism version 9.5.1 for Windows (GraphPad Software, San Diego, CA, USA) and the open-source Pingouin and Statsmodels packages in Python 3.9. The level of significance was set at $p < 0.05$.

3 Results

3.1 Athlete Travel and Sleep Interest Group (ATSIG) Experts' Answer Rate

Experts' answer rate was 100% (20/20) at T1 for both the sleep and jet lag survey and 95% (19/20) at T2 for both the sleep and jet lag survey. Given the high intra-rater reliability observed between the two time points (see below), only the data from T1 were used for all subsequent analyses.

3.2 Inter-Rater Reliability and Intra-Rater Agreement

Inter-rater reliability was classified as *minimal* (0.21–0.39) for each LLM, except for Google Bard for the sleep survey (Fleiss' Kappa: 0.17, *none*). Table 3 shows the Fleiss' Kappa values for inter-rater reliability for each LLM in the sleep and jet lag surveys.

The intra-rater agreement was always classified, on average, as *high* or *very high* and a minimum of 52.6% of experts (10/19) always reached a JSI ≥ 0.76 (*very high*) for each LLM for both surveys. The JSI values for intra-rater agreement for each expert and LLM, and for both the sleep and the jet lag surveys, are reported in Tables 4 and 5.

Table 3 Fleiss' Kappa and Jaccard Similarity Index values for inter-rater reliability and agreement, respectively, for each large language model (LLM) in sleep and jet lag survey

LLM, survey	Fleiss' Kappa
ChatGPT-3.5, sleep	0.22, minimal
Bard, sleep	0.17, none
ChatGPT-4, sleep	0.24, minimal
ChatGPT-3.5, jet lag	0.26, minimal
Bard, jet lag	0.22, minimal
ChatGPT-4, jet lag	0.26, minimal

Table 4 Jaccard Similarity Index values for intra-rater agreement for each large language model (LLM) in the sleep survey

	ChatGPT-3.5	Google Bard	ChatGPT-4
Expert 1	0.78, very high	0.76, very high	0.68, high
Expert 2	0.88, very high	0.92, very high	0.97, very high
Expert 3	0.80, very high	0.92, very high	0.76, very high
Expert 4	0.43, moderate	0.65, high	0.80, very high
Expert 5	0.47, moderate	0.60, high	0.73, high
Expert 6	–	–	–
Expert 7	0.71, high	0.86, very high	0.47, moderate
Expert 8	0.83, very high	0.53, high	0.81, very high
Expert 9	0.65, high	0.69, high	0.88, very high
Expert 10	0.68, high	0.95, very high	0.53, high
Expert 11	0.87, very high	0.92, very high	0.95, very high
Expert 12	0.95, very high	0.95, very high	0.95, very high
Expert 13	0.83, very high	0.62, high	0.95, very high
Expert 14	0.95, very high	0.88, very high	0.97, very high
Expert 15	0.92, very high	0.52, high	0.77, very high
Expert 16	0.65, high	1.0, very high	0.92, very high
Expert 17	0.62, high	0.85, very high	0.76, very high
Expert 18	0.74, high	0.50, moderate	0.55, high
Expert 19	0.27, moderate	0.73, high	0.64, high
Expert 20	0.78, very high	0.74, high	0.74, very high
Average	0.73, high	0.77, very high	0.78, very high
Low (*n*, %)	0, 0%	0, 0%	0, 0%
Moderate (*n*, %)	3, 15.8%	1, 5.3%	1, 5.3%
High (*n*, %)	6, 31.6%	8, 42.1%	5, 26.3%
Very high (*n*, %)	10, 52.6%	10, 52.6%	13, 68.4%

Data from Expert 6 are missing in the follow-up survey (T2)

3.3 Content Validity of LLMs

CVR was ≥ 0.5 in 70, 60 and 90% of the answers on sleep for ChatGPT-3.5, Google Bard and ChatGPT-4, respectively, with a CVI consistently higher than 0.5 for each LLM. Conversely, the minimum CVR value was reached only in Q1 for ChatGPT-3.5 (10%) and never for Google Bard (0%) for the jet lag survey. Only the answers provided by ChatGPT-4 on jet lag were rated as valid by the experts (100%; CVI: 0.68). OSM Sources 3a, 3b and 3c show the CVR for each of the ten questions of the sleep survey for ChatGPT-3.5, Google Bard, and ChatGPT-4, respectively. Additionally, OSM Sources 4a, 4b and 4c show the CVR for the ten questions on jet lag for each LLM.

Content validity significantly varied by LLM and survey. CVRs were higher for the sleep survey than for the jet lag survey for ChatGPT-3.5 ($p = 0.048$, ES: 1.3, *large*) and Google Bard ($p = 0.0004$, ES: 2.8, *very large*), but not for ChatGPT-4. There were no inter-LLMs differences in CVR for the sleep survey. However, for the jet lag survey, the CVR was significantly lower for Google Bard compared both with ChatGPT-3.5 ($p = 0.0073$, ES: 1.6, *large*) and

Table 5 Jaccard Similarity Index values for intra-rater agreement for each large language model (LLM) in the jet lag survey

	ChatGPT-3.5	Google Bard	ChatGPT-4
Expert 1	0.80, very high	0.92, very high	0.77, very high
Expert 2	0.29, moderate	0.27, moderate	0.0, low
Expert 3	0.60, high	0.81, very high	0.92, very high
Expert 4	0.86, very high	0.69, very high	0.88, very high
Expert 5	0.78, very high	0.95, very high	0.86, very high
Expert 6	0.83, very high	0.88, very high	0.36, moderate
Expert 7	0.88, very high	0.88, very high	0.92, very high
Expert 8	0.93, very high	0.88, very high	1.0, very high
Expert 9	–	–	–
Expert 10	0.68, high	0.62, high	0.92, very high
Expert 11	0.29, moderate	0.60, high	0.86, very high
Expert 12	0.88, very high	0.85, very high	0.98, very high
Expert 13	0.88, very high	1.0, very high	0.77, very high
Expert 14	0.85, very high	0.71, high	0.52, moderate
Expert 15	0.35, moderate	0.69, high	0.0, low
Expert 16	0.97, very high	0.79, very high	0.92, very high
Expert 17	0.92, very high	0.88, very high	0.74, high
Expert 18	0.29, moderate	0.80, very high	0.31, moderate
Expert 19	0.69, high	0.25, low	0.78, very high
Expert 20	0.60, high	0.69, high	0.69, high
Average	0.71, high	0.75, high	0.70, high
Low (*n*, %)	0, 0%	1, 5.3%	2, 10.5%
Moderate (*n*, %)	4, 21.1%	1, 5.3%	3, 15.8%
High (*n*, %)	4, 21.1%	5, 26.3%	2, 10.6%
Very high (*n*, %)	11, 57.8%	12, 63.1%	12, 63.1%

Data from Expert 9 are missing in the follow-up survey (T2)

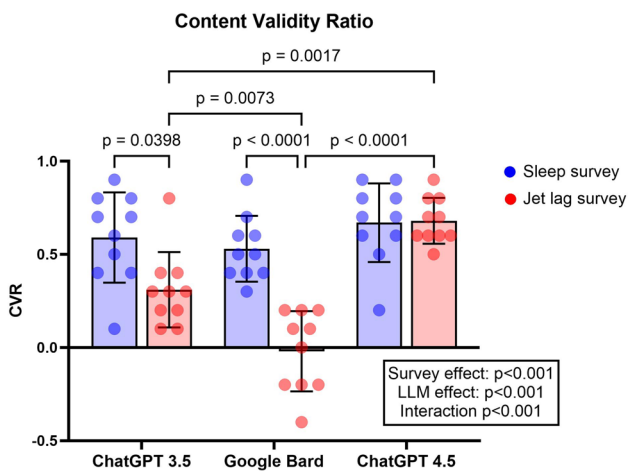

Fig. 1 Histograms with single data, mean ± SD of CVR (content validity ratio) for ChatGPT-3.5, Google Bard and ChatGPT-4. Blue histogram and circles: CVR for the sleep survey; Red histogram and circles: CVR for the jet lag survey. Results of the two-way ANOVA and Bonferroni's multiple comparisons, with associated *p*-values, are presented

ChatGPT-4 ($p < 0.0001$, ES: 4.1, *very large*). Furthermore, ChatGPT-4 reached a higher CVR than Chat-GPT3.5 for the jet lag survey ($p = 0.0017$, ES: 2.2, *very large*). Figure 1 shows histograms with mean ± SD of CVR for each LLM and both surveys.

3.4 Differences Among LLMs

The Friedman test highlighted that the ATSIG experts" scores for the sleep survey did not significantly vary among LLMs except for Q1 ($p = 0.040$), Q2 ($p = 0.007$, with Bard < ChatGPT-3.5 ($p = 0.04$, ES: 0.62, moderate)), and Q6 ($p = 0.033$). Conversely, significantly large differences among LLMs were observed in the ratings for the jet lag survey. Specifically, Google Bard was always significantly inferior to ChatGPT-4 for the responses provided for questions from Q1 to Q8, with *small* (only Q7) to *moderate* effect sizes, and to ChatGPT-3.5 only for Q1 ($p = 0.004$; ES: 0.73, moderate). Furthermore, a significant Friedman test was also reached for Q9 ($p = 0.01$) and Q10 ($p = 0.02$) with no post hoc significant differences. The median and IQR of raters' scores for each question and each LLM for the sleep and jet lag survey are graphically shown in Figs. 2 and 3, whereas OSM Resources 5 and 6 show the results of the Friedman test, multiple comparisons, and ES for significant outcomes.

3.5 Reasons for the Inappropriateness of the LLM Answers

The reasons for the inappropriateness of the answers were similar among the LLMs for the sleep survey (χ^2: 12.6, $p = 0.40$, ns) with "too few information" (ChatGPT-3.5: 19%, Google Bard: 22%, ChatGPT-4: 24%) and "other reasons" (ChatGPT-3.5: 31%, Google Bard: 25%, ChatGPT-4: 22%) being the categories most frequently selected by the ATSIG members (Fig. 4, upper panel). Among the open-text explanations provided under "other reasons", common themes included overly generic answers, lack of sport-specific context, responses were considered vague, or responses perceived as overly confident without sufficient evidence. Conversely, the cause of inappropriateness of the responses varied significantly among LLMs for the jet lag survey (χ^2: 39.7, $p < 0.0001$): "too few information" was the most frequently selected category by the ATSIG experts, especially for Google Bard (39%) compared to ChatGPT-3.5 (30%) and ChatGPT-4 (28%). In addition, Google Bard was identified as the LLM with the highest number of "clear mistakes" in the answers text (21%) and "language issues" (6%) compared to ChatGPT-3.5 (13 and 2%) and ChatGPT-4 (6 and 0%; Fig. 4, lower panel). Lastly, it is noteworthy that "language issues" were never identified by the experts as a cause of inappropriate responses for ChatGPT-4 in the sleep and jet lag surveys (0%). Furthermore, ChatGPT-4 always

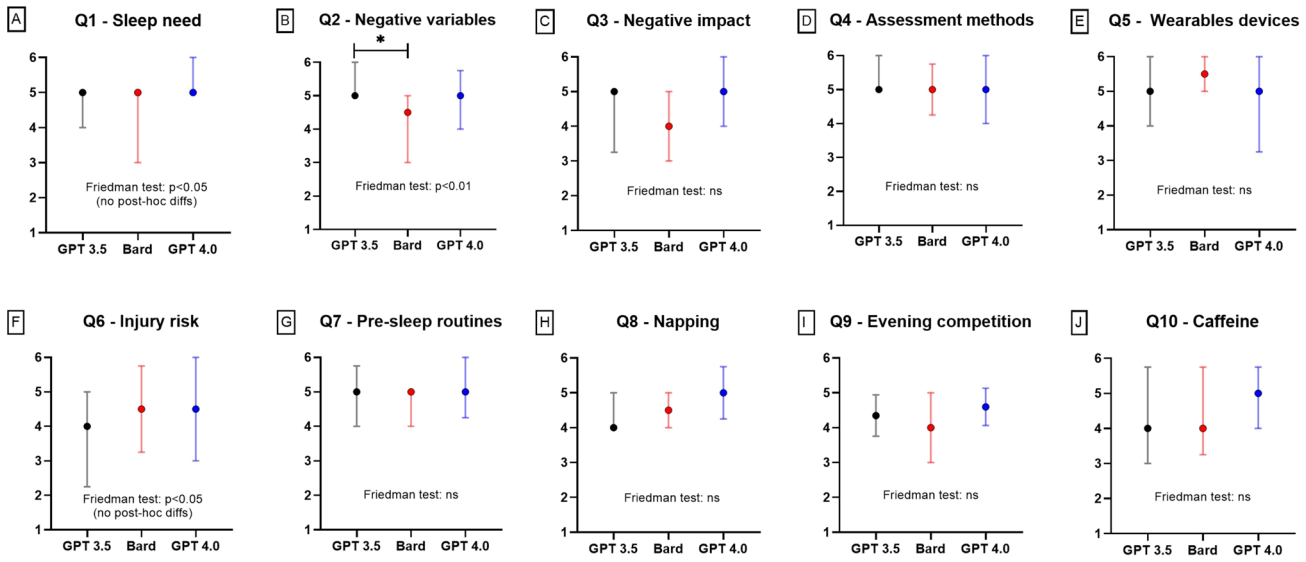

Fig. 2 Median and interquartile range (IQR) of raters' scores for each question (upper panels from **A** to **E** for Q1–Q5 and lower panels from **F** to **J** for Q6–Q10) and each LLM (ChatGPT-3.5: black lines; Google Bard: red lines; ChatGPT-4: blue lines) for the sleep survey. Legend: $*p<0.05$

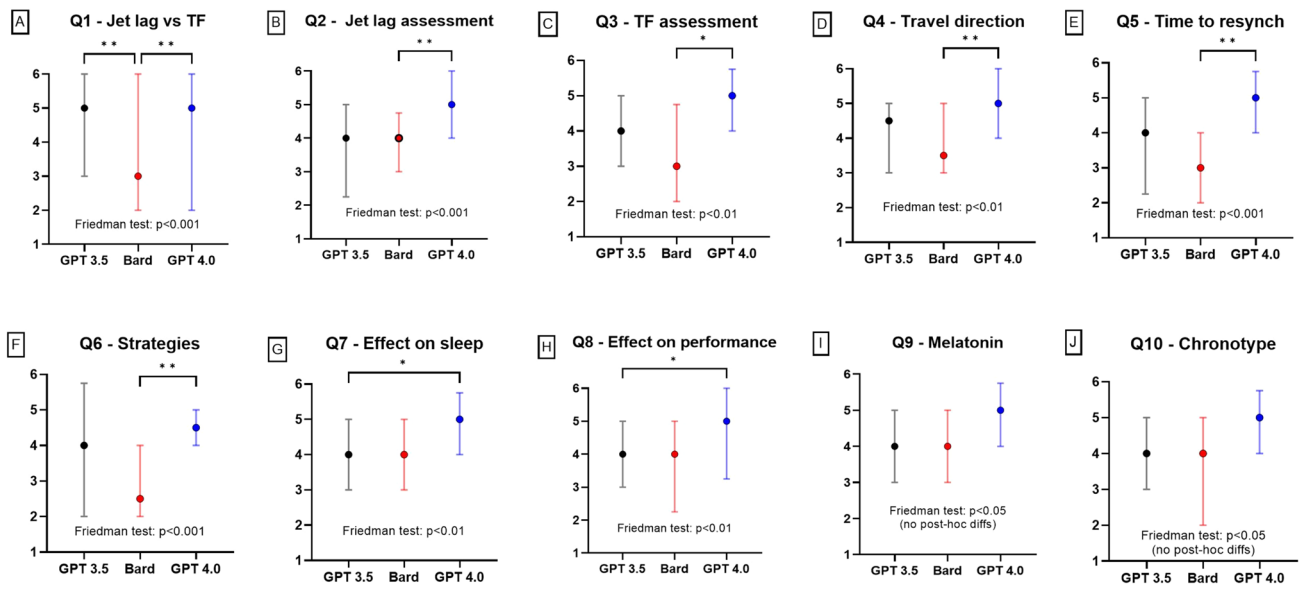

Fig. 3 Median and interquartile range (IQR) of raters' scores for each question (upper panels from **A** to **E** for Q1–Q5 and lower panels from **F** to **J** for Q6–Q10) and each LLM (ChatGPT-3.5: black lines; Google Bard: red lines; ChatGPT-4: blue lines) for the jet lag survey. Legend: $*p<0.05$; $**p<0.01$

received the highest scores of appropriateness (45% for both surveys) compared to ChatGPT-3.5 and Google Bard for sleep (33 and 36%) and for jet lag (30 and 20%).

3.6 Preferred LLM

The understandability, clarity, professionality and length of the answers were similar among ChatGPT-3.5, Google Bard and ChatGPT-4 for the sleep survey. However, for the jet lag survey, Google Bard received significantly lower ratings compared to ChatGPT-4 for each of the above-mentioned characteristics (OSM Resource 7). ChatGPT-4 was the preferred LLM by 40 and 85% of the experts for the sleep and jet lag survey, respectively, whereas Google Bard was selected as the preferred LLM only by one expert (5%) for the jet lag survey (OSM Resource 8).

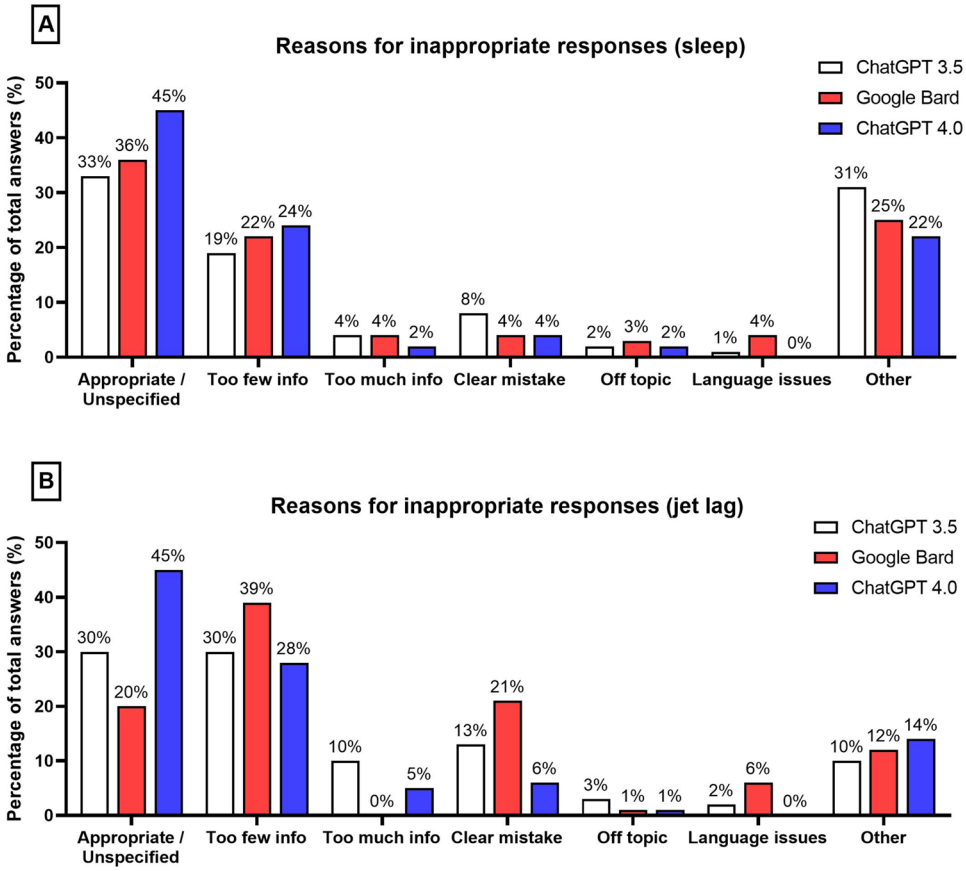

Fig. 4 Histograms (white: ChatGPT-3.5, red: Google Bard; blue: ChatGPT-4) with the percentage distribution of the reasons for inappropriate responses in the sleep (upper panel **A**) and jet lag (lower panel **B**) surveys

3.7 Risks, Advantages, Expectations and Attitude

None of the ATSIG experts reported that integrating LLMs in sports and sports science could represent a significant risk. However, 20 and 40% of them identified this integration as a moderate risk concerning the topics of sleep and jet lag, respectively. Nevertheless, 65% of the expert panel perceived the use of LLMs by coaches and athletes as a significant advantage for retrieving information on athletes' sleep, and 40% perceived it as a moderate advantage for jet lag (OSM Resource 9, upper panel).

ATSIG members' expectations of the performance of the three LLMs on the sleep topic were met (median: 5.0, IQR: 4.0–5.7) and were significantly higher compared to the jet lag topic (median: 4.0, IQR: 3.0–4.0; $p = 0.0009$, ES: 0.73, *moderate*). Lastly, experts' attitudes toward integrating LLMs in sports and sports science were positive, with only 20 and 30% of experts reporting concerns (scores ≤ 3.0) about sleep and jet lag topics, respectively (OSM Resource 9, lower panel).

4 Discussion

To the best of our knowledge, this is the first study to evaluate the performance of ChatGPT-3.5, ChatGPT-4 and Google Bard in providing evidence-based advice on sleep and jet lag specifically for athletes. Our initial hypotheses were partially confirmed. The first hypothesis predicted good content validity of the LLMs for both sleep and jet lag topics; however, this was true only for sleep but not uniformly for jet lag, where only ChatGPT-4 showed valid content. Key findings include: (1) the inter-rater reliability was low among the experts across all LLMs, whereas intra-rater agreement was high, suggesting that individual experts were consistent in their evaluations over time; (2) ChatGPT-4 achieved the highest CVR for sleep and was the only model with a valid CVR for jet lag. In contrast, Google Bard had the lowest CVR for jet lag, highlighting significant discrepancies in performance; (3) Google Bard was frequently criticized for "too few information", "clear mistakes" and "language issues", especially in the context of jet lag. Overall, ChatGPT-4 emerged as the most appropriate model, receiving no critiques for language issues and providing the most reliable responses.

4.1 ATSIG Experts' Reliability and Expertise

In the present study, we could not calculate the traditional Intraclass Correlation Coefficient (ICC) to assess inter- and intra-rater reliability. The ICC is typically used to assess intra- and inter-rater reliability in evaluating the same set of subjects or measurements. However, when the ratings from each rater are identical (either between the two raters or within the same rater at two time points) and constant (the same rating for all measures), the ICC may not be appropriate. Specifically, for ICC type 3 (ICC (3,1)), which is used when the same fixed set of raters rates each measure, the variance within subjects becomes zero with identical ratings. This results in an ICC (3,1) equal to zero due to the lack of variability to measure agreement or reliability among the raters, despite the perfect agreement between them. The ICC (3,1) formula is as follows:

$$\mathrm{ICC}_{(3,1)} = \frac{\mathrm{MS_B} - \mathrm{MS_E}}{\mathrm{MS_B} + (k-1) * \mathrm{MS_E}}$$

$\mathrm{MS_B}$ is the mean square between raters, $\mathrm{MS_E}$ is the mean square error within rater, and k is the number of raters. If we imagine that two raters always give the same score (e.g. a score of 4 of the Likert scale) to different measures (e.g. the ten answers provided by the LLMs), the MSB is equal to zero because there is no difference between the two subjects. Further, since the ratings are identical within the same rater, the MSE is also zero. These two conditions result in a numerator equal to zero; consequently, the ICC is also zero. Therefore, we could not calculate the ICC for the above-mentioned mathematical reasons. We used Fleiss' Kappa and JSI values as an alternative in our analysis.

Beyond the issues associated with not using the ICC, we observed a high intra-rater agreement, reflecting the consistency of individual experts in applying their personal evaluation criteria over time. Conversely, the observed inter-rater reliability between the experts was low. Several factors may explain the low inter-rater reliability between experts. First, assessing LLM responses on topics like sleep and jet lag involves a significant degree of subjectivity. Experts may have varying interpretations of what constitutes an "appropriate" response based on their individual experiences, biases and areas of expertise. Second, the ATSIG comprises experts from various backgrounds (e.g., sports science, sports medicine, sports nutrition, sports psychology). This diversity means that each expert might prioritize different aspects of the responses, leading to discrepancies in their evaluations. For example, medical doctors might prioritize practical, clinically relevant information when evaluating responses. They could be more focused on whether the answers provided by the LLMs align with their clinical experiences and practical knowledge, which may lead them to rate the responses more favourably if they find the information applicable and useful in real-world scenarios. However, we were not able to statistically examine whether there were significant differences in evaluating LLMs' answers based on ATSIG members' covariates, including gender, profession, age and years of experience, due to the small number of raters in each category ($n = 3$ medical doctors, $n = 4$ professors; $n = 3$ consultants and $n = 10$ researchers). This suggests that further research is needed to confirm these findings.

Nevertheless, it is important to highlight the observed low inter-rater reliability, which carries imperative implications for practical applications, especially in fields where a multidisciplinary team of experts must collaborate and provide consistent guidance to athletes. Such inconsistencies can result in mixed messages, ultimately affecting the implementation of strategies designed to optimize athletic performance or recovery. Furthermore, if experts within the same team or organization provide conflicting advice based on their interpretations of AI-generated recommendations, this could impair decision-making processes, create confusion, and potentially undermine the athletes' trust in the guidance provided. For instance, one expert may emphasize the importance of recovery time based on their experience with jet lag. At the same time, another might prioritize hydration or light exposure strategies, leading to divergent recommendations. This underscores the need for a standardized approach or framework to harmonise expert evaluations to ensure coherent communication and unified recommendations. In addition, fostering open communication and encouraging shared decision-making among experts can help bridge gaps in interpretation, ensuring that different perspectives are integrated into a cohesive strategy. Therefore, to mitigate the effects of low inter-rater reliability, it may be necessary to implement more structured evaluation criteria or use collaborative tools such as consensus meetings, where experts discuss and align on common standards. By creating opportunities for dialogue and collaboration, experts can arrive at shared decisions that strengthen both the quality of their feedback and the reliability of the advice provided to end-users, such as athletes or coaching teams.

4.2 Sleep Versus Jet Lag

Several factors could contribute to the disparity in the appropriateness of LLM-generated information on sleep versus jet lag. Sleep in the athlete context has been extensively studied, providing a vast repository of data that LLMs can potentially access and utilize, likely contributing to the higher content validity scores for sleep-related queries. A recent systematic review by Walsh et al. [18] highlighted the large volume of high-quality research on athlete sleep and its implications for performance and recovery. In contrast, jet lag, especially in the context of athletic performance,

is less comprehensively researched, posing challenges for LLMs in generating accurate and comprehensive responses [17]. For example, a search in PubMed for the terms 'sleep' and 'athletes' yields 6,686 records, whereas using the terms 'jet lag' and 'athletes' yields only 147 records, corresponding to 2.2% of the articles on athletes' sleep [26]. Jet lag encompasses a multitude of variables, including the direction of travel, the number of time zones crossed, individual circadian timing, and varying impacts on physiological and psychological performance [17], making it a more complex topic for LLMs to address effectively compared to the relatively straightforward and well-documented aspects of sleep. Additionally, effective management of jet lag involves insights from chronobiology, sleep medicine, sports science and psychology, and the integration and synthesis of such multidisciplinary data are challenging for LLMs, which may not yet fully capture the nuanced interplay of these factors.

4.3 Differences Between LLMs

In the present study, ChatGPT-4 demonstrated superior performance compared to Google Bard in addressing complex topics such as jet lag and sleep health in athletes. This finding is consistent with a recent comparative analysis indicating ChatGPT-4 could effectively answer sleep-related questions and offered better strategies for managing jet lag than Google Bard [27]. In a separate study, the ability of ChatGPT-4 to debunk sleep-related myths across various sleep-related domains was examined, showing that its assessments were highly aligned with expert opinions, as shown by strong intraclass correlation coefficients [28]. In our study, while Google Bard attempted to provide information as an expert answer, by repeating, for example, the same sentence at the beginning of every response "As an experienced chronobiologist…", this feature can lead to misperception and randomness, particularly when users seek trustworthy guidance. Furthermore, the tendency of Google Bard to produce less accurate information in more than the preset 150 words raises concerns about its effectiveness as a tool for users seeking reliable advice on managing jet lag. This condition requires detailed and actionable recommendations. Overall, ChatGPT-4 outperformed Google Bard in delivering reliable, contextually relevant information on sleep and managing jet lag in athletes.

4.4 Strengths and Limitations

The study has several notable strengths. The two-phase approach, involving identifying relevant FAQs, the pilot testing, and the expert evaluation of LLM responses, ensured a thorough and systematic assessment. This design allowed for a detailed analysis of LLM performance across two distinct but related domains, sleep and jet lag. Importantly, the study included two time points of evaluation separated by 8 weeks. By separating the evaluations, the study ensured that the experts' assessments were based on fresh impressions rather than memory of their previous ratings, thereby enhancing the reliability of the results and minimizing recall bias [29]. The involvement of a multidisciplinary group of 20 experts, the ATSIG, with significant experience in sleep and jet lag research, provided a robust foundation for evaluating the LLM responses. The balanced sex representation and the high level of professional knowledge among the experts further strengthened the validity of the findings. The study also achieved high response rates, ensuring comprehensive data collection.

Despite the aforementioned strengths, some limitations must be acknowledged. One primary limitation is the relatively small sample size of the expert panel. While the experts were well qualified and diverse, a larger sample size would have provided more generalisable results and potentially reduced the variability in the evaluations. The static nature of the LLM versions evaluated in this study represents a significant limitation. The research focused on assessing the capabilities of ChatGPT-3.5, ChatGPT-4 and Google Bard at specific points in time, capturing only a snapshot of their performance under those conditions. However, LLMs constantly develop, with frequent updates and improvements that can significantly enhance their capabilities, accuracy, and functionality. Consequently, the findings of this study may become outdated as newer versions of these models are released. For instance, Google Bard, one of the models evaluated in this study, no longer exists and has been replaced by Gemini, which was released in December 2023. Similarly, ChatGPT-3.5 has been supplanted by the more advanced ChatGPT-4 in May 2024. Furthermore, while ChatGPT-4 was initially available only through a paid subscription, it is now freely available, significantly broadening its user base. These developments underscore the importance of continuous evaluation and adaptation in research involving rapidly evolving technologies like LLMs. Further, the study did not fully explore the potential real-world applicability and integration of LLM-generated advice in athletic settings. While evaluating content validity and reliability is crucial, understanding how this advice is practically implemented and its impact on athletes' performance and well-being would provide a more comprehensive assessment of the usefulness of LLMs in sports science. Finally, including a non-expert control group—such as athletes or coaches—could help assess the accessibility, usability, and perceived value of LLM responses from the perspective of typical end-users, thereby offering a more comprehensive understanding of their practical applicability in real-world sports settings.

4.5 Takeaways for Using Large Language Models (LLMs) in Sport Science

Despite the impressive power of LLMs, deep learning researchers such as Yann LeCun have stated that these models have not yet achieved what is known as Artificial General Intelligence (AGI) [30], which refers to the ability to outperform human capabilities across a wide range of cognitive tasks [31]. LeCun argues that LLMs lack essential capabilities required for true intelligence, such as understanding and reasoning about the physical world. Additionally, they require an enormous volume of data to achieve high performance levels and cannot plan beyond the scenarios on which they have been trained. While LLMs are useful tools, they are not yet replacements for human expertise, especially in complex and nuanced domains like sports science. However, they can still be beneficial, particularly in low-resource settings with limited expert guidance. They can also be highly valuable in time-sensitive environments, such as on-field practice sessions or real-time decision making, even in resource-rich settings, as they can save valuable time for practitioners and experts. For example, LLMs like ChatGPT-4 can help educate athletes, coaches, and support staff on critical areas such as sleep and jet lag management. Nonetheless, their use must be approached with caution due to variability in their output. By recognizing both the strengths and limitations of LLMs and using them judiciously, stakeholders in sports science can leverage these tools to enhance performance and address challenges related to sleep and jet lag.

5 Conclusion

This study highlights the potential of LLMs, particularly ChatGPT-4, in delivering useful information on sleep for athletes. However, it also highlights significant challenges in the jet lag domain, where current LLMs often appear to fall short. Although LLMs can accelerate the process of answering questions and offering insights, it is crucial that they are carefully prompted and that experts review their outputs to ensure accuracy. However, as demonstrated by the low inter-rater reliability observed in this study, what one expert deems 'appropriate' may differ from another's assessment, and this variability raises concerns about the consistency and reliability of LLM-generated advice. Furthermore, the lack of a unified expert consensus on certain topics presents an additional challenge. To address this, future research should employ expert-driven methods, such as the Delphi consensus process, to establish a 'gold standard' for validating and refining LLM outputs. Based on the findings presented in this study, the answer to the question posed in the title—"Can we trust them?"—is partially affirmative, with important caveats that need to be acknowledged. LLMs can be valuable as part of a larger, expert-driven process rather than as standalone decision makers.

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Athlete Travel and Sleep Interest Group (ATSIG): Juan-Manuel Alonso, Sports Medicine Department Aspetar Hospital, Doha, Qatar; Amy Bender, University of Calgary, Calgary, Alberta, Canada; Michelle Biggins, Sport Ireland Institute, National Sports Campus, Abbotstown, Dublin, Ireland; Jonathan Charest, Université Laval, École de Psychologie, Québec, Canada; Michiko Dohi, College of Sport and Wellness, Rikkyo University, Tokyo, Japan; Aleksandra Filip-Stachnik: 1. Institute of Sports Science, Jerzy Kukuczka Academy of Physical Education in Katowice, Katowice, Poland; 2. Institute of Biomedical Sciences, University of Physical Education in Cracow, Cracow, Poland; Hugh Fullagar, 1. School of Sport, Exercise and Rehabilitation, University of Technology Sydney, Sydney, Australia; 2. Department of Sport Science, Reykjavik University, Iceland; 3. Aspetar Orthopaedic and Sports Medicine Hospital, Doha, Qatar; Micheal Grandner: Department of Psychiatry, University of Arizona—Tucson, Tucson, AZ USA; Luke Gupta, UK Sports Institute, Performance Innovation, The High Performance Centre, Bisham Abbey National Sports Centre, UK; Shona Halson: School of Behavioural and Health Sciences, Australian Catholic University, Brisbane, QLD, Australia, Sports Performance, Recovery, Injury and New Technologies (SPRINT) Research Centre, Australian Catholic University, Brisbane, QLD, Australia; Michele Lastella, Appleton Institute for Behavioural Science, Central Queensland University, Adelaide, SA, Australia; Dean Miller, Appleton Institute for Behavioural Science, Central Queensland University, Adelaide, SA, Australia; Mathieu Nedelec, French Institute of Sport (INSEP), Laboratory Sport, Expertise and Performance, Paris, France; Audrey Jansen van Rensburg, Section Sports Medicine, Faculty of Health Sciences, University of Pretoria, Pretoria, South Africa; Dina C. Janse van Rensburg: Section Sports Medicine and SEMLI, Faculty of Health Sciences, University of Pretoria, Pretoria, South Africa, Medical Advisory Panel, World Netball, Manchester, UK; Gregory D. Roach, Appleton Institute for Behavioural Science, Central Queensland University, Adelaide, SA, Australia; Antonia Rossiter, 1. Sport Ireland Institute National Sports Campus Dublin, Ireland; 2. Sport and Human Performance Research Centre Health Research Institute University of Limerick Limerick, Ireland; Charli Sargent, Appleton Institute for Behavioural Science, Central Queensland University, Adelaide, Australia; Sabrina Skorski, Institute for Sport and Preventive Medicine, Saarland University, Germany; Jacopo Vitale: Research Group Spine, Department of Teaching, Research and Development, Schulthess Klinik, Lengghalde 2, 8008, Zurich, Switzerland, Institute of Sports Sciences, University of Physical Culture in Cracow, 31-571 Cracow, Poland.

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Availability of data and material The datasets generated and analyzed during the current study are available from the corresponding author, Jacopo Vitale (jacopo.vitale@kws.ch), upon reasonable request.

Ethics approval As officially specified by a letter from the local Ethical Committee of the Principal Investigator (JV, dated 15/12/2023), ethical approval was not required for the present study because the research project does not fall within the scope of the Human Research Act.

Consent to participate Not applicable.

Consent for publication Not applicable.

Author contributions The study was conceptualized and designed by JV, DCJvR, and SH. JV, SH, DCJvR, AC, and AM developed the methodology, including FAQ identification and pilot testing. JV, DCJvR, and SH conducted data collection, curation, and survey administration. Statistical analyses were performed by JV, with critical input from AC and AM. The ATSIG contributed to the evaluation and interpretation of the LLM responses and provided critical insights during manuscript preparation. The initial draft of the manuscript was written by JV, with substantial revisions contributed by AM, DCJvR, SH, and AC. All authors, including ATSIG members, reviewed and critically revised the manuscript, providing intellectual content. All authors approved the final version of the manuscript. JV is the guarantor of this work.

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Authors and Affiliations

Jacopo Vitale[1,2] · Alan McCall[3,4,5] · Andrea Cina[1,6] · Athlete Travel, Sleep Interest Group (ATSIG) · Dina C. Janse van Rensburg[7,8] · Shona Halson[9,10]

✉ Jacopo Vitale
jacopo.vitale@kws.ch

1 Schulthess Klinik, Lengghalde 2, 8008 Zurich, Switzerland

2 Institute of Sports Sciences, University of Physical Culture in Cracow, 31-571 Cracow, Poland

3 School of Applied Sciences, Edinburgh Napier University, Edinburgh, UK

4 FIFA Medical, Fédération Internationale de Football Association, Zurich, Switzerland

5 School of Sport, Exercise and Rehabilitation, University of Technology Sydney, Sydney, NSW, Australia

6 ETH Zurich, Zurich, Switzerland

7 Section Sports Medicine and SEMLI, Faculty of Health Sciences, University of Pretoria, Pretoria, South Africa

8 Medical Advisory Panel, World Netball, Manchester, UK

9 School of Behavioural and Health Sciences, Australian Catholic University, Brisbane, QLD, Australia

10 Sports Performance, Recovery, Injury and New Technologies (SPRINT) Research Centre, Australian Catholic University, Brisbane, QLD, Australia